

Fractional Structure and Möbius Transformation — Part II: The Critical-One Hypothesis and the Egg of Infinity

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Abstract

Part I captured ordinary division as a *double inversion*: the operator switches $(\div \rightarrow \times)$ while the operand inverts $(y \mapsto y^{-1})$. There we situated the transform $T : (\div, y) \mapsto (\times, y^{-1})$ inside the group of linear fractional (Möbius) maps on the projective line and highlighted the hinge role of 1.

This Part II advances the *critical-one hypothesis*: for reciprocal symmetry, the effective singular locus is not at 0 but at 1. We justify this by (i) conjugating reciprocity to a reflection in logarithmic coordinate $s = \log x$, (ii) exhibiting an I -invariant “energy” $E(x) = (\log x)^2$ minimized at $x = 1$, and (iii) deriving a projective normal form that realizes a half-twist with 1 as the mirror center. We discuss operational and philosophical consequences and outline tests for Part III.

Keywords Möbius transformation; reciprocal symmetry; critical point; logarithmic coordinate; projective line; twist phase.

1 Introduction & statement of the hypothesis

On $\mathbb{R}_{>0}$, let $I(x) = 1/x$. The map exchanges 0 and ∞ and fixes 1. We call 1 the *critical point of reciprocity* because it is the unique positive fixed point with $I'(1) = -1$; hence 1 is an orientation-reversing mirror (in $s = \log x$).

Critical-one hypothesis. Reciprocal singular effects concentrate near $x = 1$ (the mirror), not near $x = 0$.

2 Conjugacy to a reflection

Let $h(x) = \log x$, with inverse $h^{-1}(s) = e^s$. A direct computation gives

$$(h \circ I \circ h^{-1})(s) = -s.$$

Thus I is *exactly* a reflection through $s = 0$ in the s -axis. The fixed set of $-s$ is $\{0\}$, corresponding to $x = 1$. There is no higher-order defect: the linearization of I at 1 already captures the full dynamics (normal form by smooth conjugacy).

3 Invariant distance and energy around 1

Define the reciprocal-invariant radius

$$\rho(x) := |\log x| = |h(x)|.$$

Then

$$\rho(I(x)) = |\log(1/x)| = |-\log x| = \rho(x).$$

Hence the I -invariant “energy”

$$E(x) := \rho(x)^2 = (\log x)^2.$$

It is minimized uniquely at $x = 1$ and increases symmetrically for $x > 1$ and $x < 1$. Hence 1 is the *phase center* of the reciprocal symmetry; the twist index can be taken as $\phi(x) = \text{sign}(\log x) \in \{-1, +1\}$, flipping under I .

4 Projective normal form and the half-twist

On the projective line $\mathbb{P}^1$, introduce the hinge coordinate

$$u := \frac{x-1}{x+1}, \quad x = \frac{1+u}{1-u}.$$

A short calculation shows $I(x) = 1/x$ yields $u \mapsto -u$. Thus $u = 0$ (i.e. $x = 1$) is the mirror, and the action is a *half-twist* about the hinge. Any linear fractional map conjugate to I near $x = 1$ shares this local form, confirming that 1 is the structural center of reciprocal geometry.

5 Operational consequences

- **Renormalized scale.** Distances are naturally measured by $|\log x|$ from 1 (not by $|x|$ from 0). Small parameters are *multiplicative*, $x = 1 \pm \epsilon$ with $|\epsilon| \ll 1$ interpreted via $s = \log x \approx \pm \epsilon$.
- **Stability of laws.** Laws invariant under I depend on $E = (\log x)^2$, hence are even in s . Odd responses encode the twist phase $\phi = \text{sign}(s)$.
- **Measurement semantics.** “Inside/outside of scale” is judged relative to 1; inversion swaps perspectives but preserves E .

6 Math–philosophy correspondence (minimal schema)

- **Singularity at unity** ($x = 1$ fixed with $I'(1) = -1$) $\leftrightarrow$ *hinge of duality* (self/other mirror where orientation flips).
- **Invariant energy** $E = (\log x)^2 \leftrightarrow$ *auditable neutrality* (twist strength independent of side, phase carried by the sign of $\log x$).
- **Half-twist normal form** ($u \mapsto -u$) $\leftrightarrow$ *minimal twist act* (one-sided band with a single hinge).

7 Conclusion & outlook to Part III

We proved that reciprocity is conjugate to a reflection with mirror at $x = 1$, established the invariant energy E , and derived a projective normal form realizing a half-twist about the hinge. These support the *critical-one hypothesis*: the effective singular locus of reciprocal symmetry sits at 1 (the “egg of infinity”).

Part III will develop quantitative tests (e.g. invariants built from E and phase), explore couplings to linear fractional flows, and connect to measurement protocols.